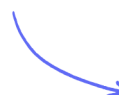


Structured Questions

Groups 1 & 2

Ionisation Energy - Groups 1 & 2 / Reactivity - Groups 1 & 2 / Oxides & Hydroxides / Group 2 Hydroxides & Sulfates / Nitrates & Carbonates / Flame Tests / Qualitative Tests

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Total Marks

/29

1 (a) This question is about Group 1 metals.

When potassium is placed into a beaker of cold water, potassium hydroxide and hydrogen are formed.

i) Write the equation for this reaction. Include state symbols.

(2)

ii) This is a redox reaction.

State which element is oxidised and which is reduced.

Justify your answer by giving the initial and final oxidation numbers of any element that changes oxidation state.

(2)

iii) The reaction of potassium with water is very vigorous and a flame is seen.

State the colour of the flame.

(1)

(5 marks)

(b) The label has come off a bottle known to contain **M**, a Group 1 metal which is stored in oil. A student carried out an experiment to determine the identity of **M**.

Procedure

Step 1

A small piece of **M** was wiped with tissue paper to remove the oil.
The piece of **M** was weighed and placed in a beaker of distilled water.

Step 2

After the reaction had finished, the contents of the beaker and washings were transferred to a 250.0 cm³ volumetric flask. The solution was made up to the mark with distilled water and mixed thoroughly.

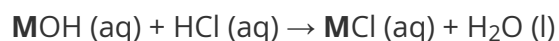
Step 3

A pipette was used to transfer 25.0 cm³ portions of this solution to conical flasks. Each portion was then titrated with hydrochloric acid of concentration 0.400 mol dm⁻³.

Results

Mass of metal, M	0.37 g
Mean titre of hydrochloric acid	12.80 cm ³

The reaction taking place is shown.



i) The indicator used was phenolphthalein. State the colour **change** at the end-point.

(2)

from to

ii) Calculate the relative atomic mass of **M** and use it to identify the Group 1 metal, **M**.

(4)

.....
.....
.....
(6 marks)

- (c) Another student repeated the experiment, using a different sample of metal **M**, but did not wipe off the oil before weighing it.

State how this would change the calculated value of the relative atomic mass of **M**. Justify your answer.

.....
.....
(2 marks)

- 2 (a) Magnesium oxalate, MgC_2O_4 , decomposes on heating to form magnesium carbonate and carbon monoxide.



A 6.0 g sample of magnesium oxalate was heated for three minutes but the decomposition was only 70% complete.

Calculate the total mass of solid that remains.

(4 marks)

- (b) In practice, magnesium carbonate also decomposes on stronger heating.

Describe and explain the trend in the thermal decomposition of Group 2 carbonates.

(3 marks)

- (c) Suggest why a pure sample of magnesium carbonate will **not** be produced from the decomposition of magnesium oxalate even if the sample is heated for longer.

(1 mark)

- 3 (a)** Magnesium ethanedioate (MgC_2O_4) decomposes on gentle heating to form magnesium carbonate and carbon monoxide.



i) State why the thermal decomposition of magnesium ethanedioate should be carried out in a fume cupboard.

(1)

ii) After heating a 6.0 g sample of magnesium ethanedioate for three minutes, the decomposition was 70% complete.

Calculate the total mass of the solid mixture that remains.

(4)

(5 marks)

- (b)** Magnesium carbonate undergoes thermal decomposition at a higher temperature than magnesium ethanedioate.



Explain the trend in the thermal decomposition of Group 2 carbonates going down the group.

(3 marks)